

Yiran Hu

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SUMMARY

Ph.D. researcher who builds, evaluates, and optimizes LLM coding agents, spanning agent system design, agent skills, and benchmarks. First-author work on test-driven repository-level code generation (ISSRE 2026) and agent cost efficiency analysis; co-author of REPOCOD (ACL 2025) and SELP (ICRA 2025, **Best Paper Award Finalist**).

EDUCATION

Purdue University, West Lafayette, IN Aug 2023 – May 2028 (Expected)
Ph.D. in Computer Science · Advisor: [Prof. Lin Tan](#)
Tianjin University, Tianjin, China Sep 2019 – Jun 2023
B.S. in Computer Science and Technology · GPA: 89.45/100 (3.73/4.0) · Advisor: [Prof. Junjie Chen](#)

PUBLICATIONS

- [P1] **Analyzing and Mitigating Cost-Inefficient Behaviors in Coding Agents** Under Review
Yiran Hu, Shanchao Liang, Nan Jiang, Yi Wu, Lin Tan.
- [C1] **TENET: One Step Toward Test-Driven Development for Repository-Level Code Generation** ISSRE 2026
Yiran Hu, Nan Jiang, Shanchao Liang, Yi Wu, Lin Tan. [\[PDF\]](#)
- [C2] **Can Language Models Replace Programmers for Coding? REPOCOD Says ‘Not Yet’** ACL 2025
Shanchao Liang, Nan Jiang, Yiran Hu, Lin Tan. [\[PDF\]](#)
- [C3] **SELP: Generating Safe and Efficient Task Plans for Robot Agents with LLMs** ICRA 2025
Yi Wu, Zikang Xiong, Yiran Hu, Shreyash S. Iyengar, Nan Jiang, Aniket Bera, Lin Tan, Suresh Jagannathan. [\[PDF\]](#)
Best Paper Award Finalist
- [C4] **LocRDF: An Ontology-Aware Key-Value Store for Massive RDF Data** WISA 2022
Yunxiao Zhang, Jinghan Li, Xueyang Liu, Rong Cheng, Yiran Hu, Xin Wang. [\[PDF\]](#)

RESEARCH EXPERIENCE

ASSET Research Lab, Purdue University – Graduate Research Assistant Aug 2023 – Present

Analyzing and Optimizing Cost Efficiency of Coding Agents [P1, first author]

- Built a trajectory-analysis pipeline over **1,200 Claude Code and Mini-SWE-Agent** trajectories on SWE-bench Verified, identifying three cost-inefficient behaviors (**subsumed retrieval, similar script generation, test re-execution**) that appear in 79%–98% of tasks and consume up to **22.75%** of task cost.
- Normalized heterogeneous agent traces into a unified action schema and combined **rule-based and LLM-based labeling** for fine-grained behavior analysis across agent architectures.
- Evaluated **structure-aware retrieval** ([CodeGraph](#) via MCP), **agent-synthesized skills**, and developer-designed skills over **10k+ agent runs** on held-out **SWE-bench Verified and Pro** tasks, executed as **12–32 parallel Docker containers** with automatic retry (~\$5K API budget), measuring Pass@1, cost, and cost-of-pass across repeated runs.
- Designed behavior-targeted **agent skills** that cut task cost by **7.88%–41.73%** and the frequency of the cost-inefficient behaviors by up to **88.91%** while maintaining Pass@1.

TENET: Test-Driven Coding Agent for Repository-Level Code Generation [C1, first author]

- Built a test-driven coding agent combining **diversity-based test selection, a retrieval (AST) & debugging (pdb) toolset, and reflection-based refinement** for repository-level code generation.
- Achieved **69.08%** Pass@1 on REPOCOD and **81.77%** on RepoEval with Claude Sonnet 4, beating the strongest agent baselines (incl. OpenHands, SWE-Agent) by **9.49** and **2.17** pp (**20.10** and **6.97** pp with DeepSeek-V3), while using **3× fewer input tokens** than SWE-Agent.
- Designed a test harness that exposes only a small, diverse set of developer-written tests as executable specifications and holds out the remaining tests to guard against **test overfitting and reward hacking**; it is TENET’s most impactful component (–**16.12** pp Pass@1 when removed) and lifts SWE-Agent by **+13.98** pp.

REPOCOD: Repository-Level Code Generation Benchmark [C2]

- Curated **980 tasks from 11 popular Python repositories** with an average of 314 developer-written tests per task.
- Built **repository-specific Docker environments** for reproducible, **execution-based evaluation** that double as **verifiable-reward environments** for agent training.
- Ran the commercial-LLM evaluations with **RAG** (including BM25); none of the 10 LLMs exceeded 30% Pass@1.

SELP: Safe and Efficient LLM Planners for Robot Agents [C3]

- Implemented the **equivalence-voting mechanism** for LLM-generated Linear Temporal Logic (LTL) specifications, raising translation accuracy from 88.4% to **98.0%** (drone navigation) and 87.4% to **95.2%** (tabletop manipulation).
- Built two datasets for drone navigation and tabletop manipulation and contributed to **supervised fine-tuning (SFT)** and evaluation; SELP improved safety rate by 10.8% (navigation) and 20.4% (manipulation) over prior planners.

Software Analysis and Intelligence Lab, Tianjin University – Undergraduate Researcher Nov 2021 – May 2023

- Ported the LEMON **deep-learning library testing** tool from **TensorFlow** to MindSpore, uncovering **8 bugs**.
- Applied **differential testing** to the LLVM and TVM compilers.
- Optimized ResNet-50 with TVM graph- and operator-level tuning, reducing inference latency by 14.4 ms.

COURSE PROJECTS

Test-Time Compute Scaling for LLM Math Reasoning

Spring 2026

- Ran experiments comparing two test-time scaling strategies, **Self-Consistency** and **PRM-weighted Best-of-N**, on **MATH-500** using Llama-3.2-1B/3B-Instruct with a Llama-3.1-8B process reward model at budgets $n=1-64$.
- Found that at $n=64$, correct answers exist in 73%–81% of Self-Consistency candidate pools, but majority voting often fails to pick them (final accuracy: 38.0%–56.2%); PRM-weighted Best-of-N beats Self-Consistency by **+5.4 pp** on Llama-3.2-1B.

INDUSTRY EXPERIENCE

Zhongke Wenge Science and Technology Co., Ltd., Beijing, China

May 2022 – Aug 2022

Algorithm Engineer Intern, Research & Development Center

- Diagnosed PaddleOCR failure modes on COCO-Text and SVT: background interference, dense text, curved layouts.
- Integrated SAST detection with SVTR recognition (ICDAR2015), improving text detection accuracy by **27.8%**; added a recognition module for previously incompatible detectors, improving accuracy by **15.4%**.

TEACHING

Teaching Assistant, Purdue University

Aug 2023 – Present

CS 176 Data Engineering in Python · CS 251/253 Data Structures & Algorithms · CS 571 Artificial Intelligence

HONORS & AWARDS

Best Paper Award Finalist , IEEE International Conference on Robotics and Automation (ICRA)	2025
Outstanding Graduate, Tianjin University	2023
First Prize, Challenge Cup National College Student Sci-Tech Competition (Tianjin University round)	2022
First Class Scholarship, Tianjin University	2021

SKILLS

Languages: Python, C++

LLM Agents: Claude Code, Codex, Google ADK, Mini-SWE-Agent, OpenHands, MCP, Agent Skills

LLM & ML: Anthropic/OpenAI APIs & SDKs, Hugging Face, PyTorch, TensorFlow, Fine-Tuning (SFT), RAG

Evaluation: SWE-bench, Benchmark Design, Agent Trajectory Analysis, Agent Efficiency Analysis

Infrastructure: Docker, Linux, Git, Pytest, Tree-sitter